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Bayesian Networks

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Probability Review: Uncertainty

Let action A_t = leave for airport t minutes before flight

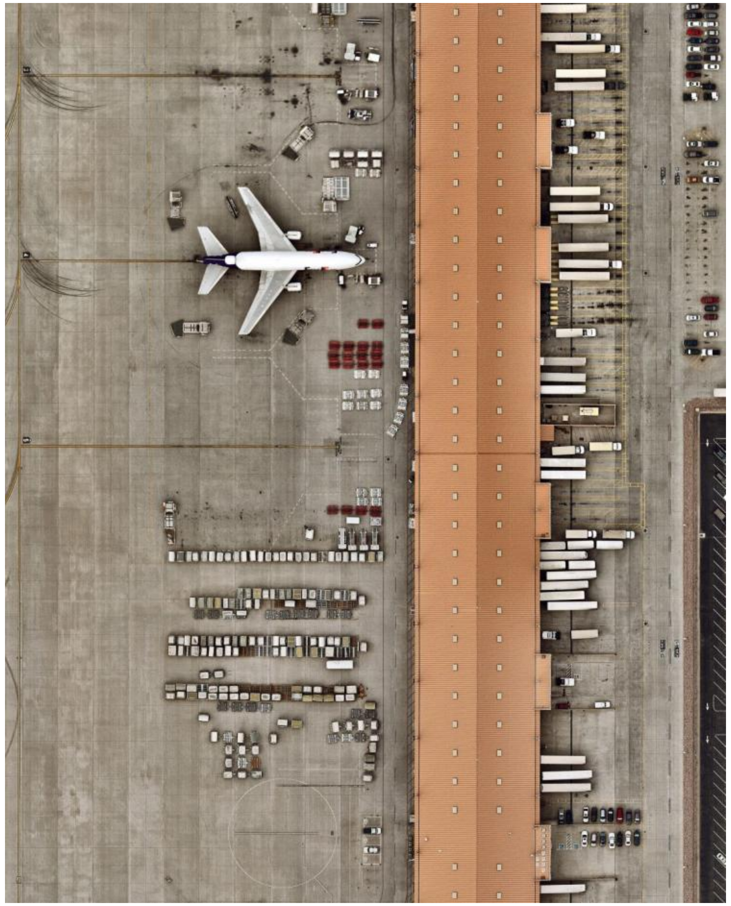
- Will A_t get me there on time?

Problems:

- **Partial observability** (road state, other drivers' plans, etc.)
- **Uncertainty** in action **outcomes** (flat tire, etc.)
- **Complexity** of **modeling** and **predicting** traffic

Hence a purely logical approach either

- Risks falsehood: "A_{25} will get me there on time," or
- Leads to conclusions that are too weak for decision making:
  - A_{25} will get me there on time if there's no accident on the bridge and it doesn't rain and my tires remain intact, etc., etc.
  - A_{1440} might reasonably be said to get me there on time but I'd have to stay overnight in the airport



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Making decisions under uncertainty

Suppose the agent believes the following:

$P(A_{25}$ gets me there on time$) = 0.04$
$P(A_{90}$ gets me there on time$) = 0.70$
$P(A_{120}$ gets me there on time$) = 0.95$
$P(A_{1440}$ gets me there on time$) = 0.9999$

Which **action** should the **agent choose**?

- Depends on **preferences** for missing flight vs. time spent waiting
- Encapsulated by a ***utility function***

The agent should choose the action that **maximizes** the ***expected utility***:

$$P(A_t \text{ succeeds}) * U(A_t \text{ succeeds}) + P(A_t \text{ fails}) * U(A_t \text{ fails})$$

Utility theory is used to represent and infer preferences

Decision theory = probability theory + utility theory

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Where do probabilities come from?

Frequentism

- Probabilities are **relative frequencies**
- For example, if we toss a coin many times, P(heads) is the proportion of the time the coin will come up heads

Subjectivism (Bayesianism)

- Probabilities are **degrees of belief**
- But then, how do we assign belief values to statements?

Random variables

We describe the (uncertain) state of the world using *random variables*

- Denoted by capital letters
- **R**: *It will rain tomorrow*
- **W**: *Weather condition*
- **D**: *Outcome of rolling two dice*
- **S**: *Speed of my car (in KPH)*

Just like variables in CSP's, random variables take on values in a *domain*

- Domain values must be mutually exclusive and exhaustive
- **R** in {True, False}
- **W** in {Sunny, Cloudy, Rainy, Snow}
- **D** in {(1,1), (1,2), … (6,6)}
- **S** in [0, 260]

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Random Variables

Definition: A *random variable* is a **function** from the sample space of an experiment to the set of real numbers. That is, a random variable assigns a real number to each possible outcome.

- A random variable is a function. It is **not a variable**, and it is **not random**!
- Think of the event as occurring or not occurring.

Example: Suppose that a coin is flipped three times. Let $X(t)$ be the random variable that equals the number of heads that appear when t is the outcome.

Then $X(t)$ takes on the following values:

$X(HHH) = 3$, $X(TTT) = 0$,
$X(HHT) = X(HTH) = X(THH) = 2$,
$X(TTH) = X(THT) = X(HTT) = 1$.

Each of the eight possible outcomes has probability 1/8.

So, the distribution of $X(t)$ is $p(X = 3) = 1/8$, $p(X = 2) = 3/8$, $p(X = 1) = 3/8$, and $p(X = 0) = 1/8$.

Events

Probabilistic statements are defined over *events*, or sets of world states

- *"It will rain tomorrow"*
- *"The weather is either cloudy or snowy"*
- *"The sum of the two dice rolls is 11"*
- *"My car is going between 50 and 90 kilometers per hour"*

Events are described using propositions:

- R = True
- W = "Cloudy" $\vee$ W = "Snowy"
- D $\in$ {(5,6), (6,5)}
- $50 \leq S \leq 90$

Notation: P(A) is the probability of the set of world states in which proposition A holds

- P(X = x), or P(x) for short, is the probability that random variable X has taken on the value x

Atomic events

Atomic event: a complete specification of the state of the world, or a complete assignment of domain values to all random variables

- Atomic events are mutually exclusive and exhaustive

E.g., if the world consists of only two Boolean variables *Cavity* and *Toothache*, then there are 4 distinct atomic events:

$Cavity = false \wedge Toothache = false$
$Cavity = false \wedge Toothache = true$
$Cavity = true \wedge Toothache = false$
$Cavity = true \wedge Toothache = true$

Joint probability distributions

A **joint distribution** is an assignment of probabilities to every possible atomic event

Atomic event	**P**
Cavity = false $\wedge$ *Toothache = false*	0.8
Cavity = false $\wedge$ *Toothache = true*	0.1
Cavity = true $\wedge$ *Toothache = false*	0.05
Cavity = true $\wedge$ *Toothache = true*	0.05

- From the axioms of probability it follows that the probabilities of all possible atomic events must sum to 1.

Joint probability distributions

Suppose we have a **joint *distribution*** $P(X_1, X_2, \ldots, X_n)$ of n random variables with domain sizes d.

- What is the size of the **probability table**?
- **Impossible** to write out completely for all but the smallest distributions

Notation:

- $P(X = x)$ is the probability that random variable X takes on value x
- $P(X)$ is the ***distribution*** of probabilities for all possible values of X

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Marginal probability distributions

Suppose we have the **joint distribution** P(X,Y) and we want to find the ***marginal distribution*** P(Y)

P(Cavity, Toothache)	
Cavity = false ∧ *Toothache = false*	0.8
Cavity = false ∧ *Toothache = true*	0.1
Cavity = true ∧ *Toothache = false*	0.05
Cavity = true ∧ *Toothache = true*	0.05

P(Cavity)	
Cavity = false	?
Cavity = true	?

P(Toothache)	
Toothache = false	?
Toothache = true	?

Marginal probability distributions

Suppose we have the **joint distribution** P(X,Y) and we want to find the ***marginal distribution*** P(Y)

P(Cavity, Toothache)	
Cavity = false ∧ *Toothache = false*	0.8
Cavity = false ∧ *Toothache = true*	0.1
Cavity = true ∧ *Toothache = false*	0.05
Cavity = true ∧ *Toothache = true*	0.05

P(Cavity)	
Cavity = false	0.9
Cavity = true	0.1

P(Toothache)	
Toothache = false	0.85
Toothache = true	0.15

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Marginal probability distributions

Suppose we have the **joint distribution** P(X,Y) and we want to find the ***marginal distribution*** P(X)

$$P(X = x) = P\big((X = x \wedge Y = y_1) \vee \ldots \vee (X = x \wedge Y = y_n)\big)$$

$$= P\big((x, y_1) \vee \ldots \vee (x, y_n)\big) = \sum_{i=1}^{n} P(x, y_i)$$

General rule: to find P(X = x), **sum** the **probabilities** of **all atomic events** where X = x.

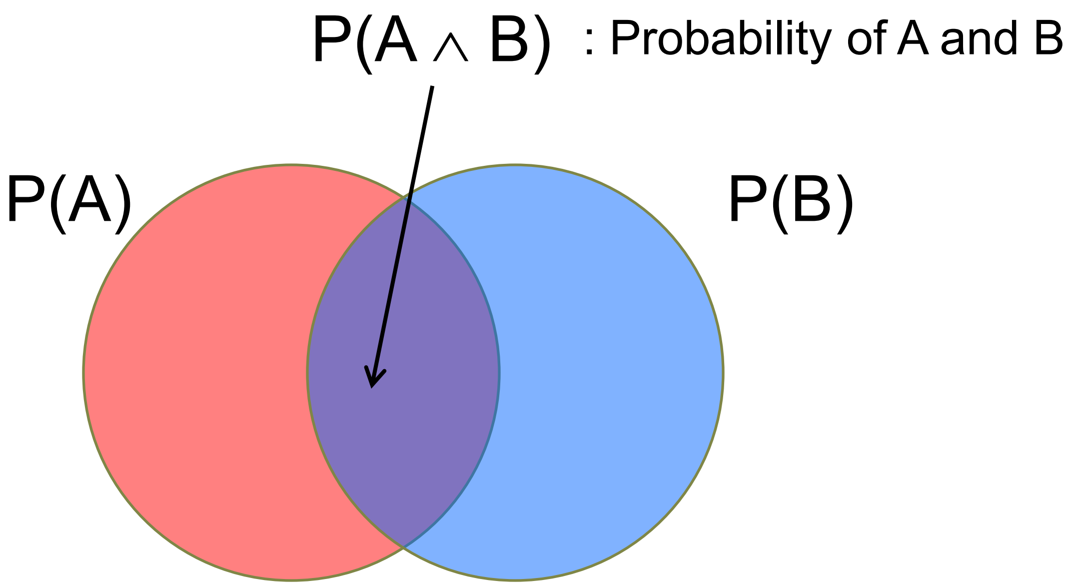
Conditional probability

Probability of cavity given toothache:

P(*Cavity = true* | *Toothache = true*)

For any two events A and B,

$$P(A \mid B) = \frac{P(A \wedge B)}{P(B)} = \frac{P(A, B)}{P(B)}$$



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Conditional probability

P(Cavity, Toothache)	
$Cavity = false \wedge Toothache = false$	0.8
$Cavity = false \wedge Toothache = true$	0.1
$Cavity = true \wedge Toothache = false$	0.05
$Cavity = true \wedge Toothache = true$	0.05

P(Cavity)	
$Cavity = false$	0.9
$Cavity = true$	0.1

P(Toothache)	
$Toothache = false$	0.85
$Toothache = true$	0.15

What is $P(Cavity = true \mid Toothache = false)$?

What is $P(Cavity = false \mid Toothache = true)$?

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Conditional probability

P(Cavity, Toothache)	
Cavity = false ∧ *Toothache = false*	0.8
Cavity = false ∧ *Toothache = true*	0.1
Cavity = true ∧ *Toothache = false*	0.05
Cavity = true ∧ *Toothache = true*	0.05

P(Cavity)	
Cavity = false	0.9
Cavity = true	0.1

P(Toothache)	
Toothache = false	0.85
Toothache = true	0.15

What is P(*Cavity = true* | *Toothache = false*)?

P(*Cavity = true* ∧ *Toothache = false*) / P(*Toothache = false*)

= 0.05 / 0.85 = 0.059

What is P(*Cavity = false* | *Toothache = true*)?

P(*Cavity = false* ∧ *Toothache = true*) / P(*Toothache = true*)

= 0.1 / 0.15 = 0.667

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Conditional distributions

A **conditional distribution** is a distribution over the values of one variable given fixed values of other variables

P(Cavity, Toothache)	
Cavity = false $\wedge$ *Toothache = false*	0.8
Cavity = false $\wedge$ *Toothache = true*	0.1
Cavity = true $\wedge$ *Toothache = false*	0.05
Cavity = true $\wedge$ *Toothache = true*	0.05

P(Cavity | Toothache = true)	
Cavity = false	0.667
Cavity = true	0.333

P(Cavity | Toothache = false)	
Cavity = false	0.941
Cavity = true	0.059

P(Toothache | Cavity = true)	
Toothache= false	0.5
Toothache = true	0.5

P(Toothache | Cavity = false)	
Toothache= false	0.889
Toothache = true	0.111

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Normalization trick

To get the whole **conditional distribution** P(X | y) at once, select all entries in the **joint distribution matching** Y = y and **renormalize** them to **sum to one**

P(Cavity, Toothache)	
Cavity = false $\wedge$ *Toothache = false*	0.8
Cavity = false $\wedge$ *Toothache = true*	0.1
Cavity = true $\wedge$ *Toothache = false*	0.05
Cavity = true $\wedge$ *Toothache = true*	0.05

Select

Toothache, Cavity = false	
Toothache= false	0.8
Toothache = true	0.1

Renormalize (Divide by 0.8+0.1= 0.9)

P(Toothache | Cavity = false)	
Toothache= false	0.889
Toothache = true	0.111

Bayes rule

Definition of **conditional probability**: $P(A \mid B) = \dfrac{P(A,B)}{P(B)}$

Sometimes we have the conditional probability and want to obtain the **joint**:

$$P(A,B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

The chain rule:

$$P(A_1,\ldots,A_n) = P(A_1)P(A_2 \mid A_1)P(A_3 \mid A_1,A_2)\ldots P(A_n \mid A_1,\ldots,A_{n-1})$$
$$= \prod_{i=1}^{n} P(A_i \mid A_1,\ldots,A_{i-1})$$

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Bayes Rule

Rev. Thomas Bayes
(1702-1761)

The product rule gives us two ways to factor a joint distribution:

$$P(A,B) = P(A \mid B)P(B) = P(B \mid A)P(A)$$

Therefore,

$$P(A \mid B) = \frac{P(B \mid A)P(A)}{P(B)}$$

Why is this useful?

- Can get *diagnostic probability* P(cavity | toothache) from *causal probability* P(toothache | cavity)
- Can update our beliefs based on evidence
- Important tool for probabilistic inference

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Bayes Rule example

Given:

- A doctor knows that meningitis causes 50% stiff neck
- The probability of any patient having meningitis based on historical data is 1/50,000.
- According to historical data, the rate of neck stiffness for any patient is known as 1/20.

If a patient complains of stiff neck, what is the probability of menangitis?

$$P(M \mid S) = \frac{P(S \mid M)P(M)}{P(S)} = \frac{0.5 \times 1/50000}{1/20} = 0.0002$$

Bayes Rule example

Marie is getting married tomorrow, at an outdoor ceremony in the desert. In recent years, it has rained only 5 days each year (5/365 = 0.014). Unfortunately, the weatherman has predicted rain for tomorrow. When it actually rains, the weatherman correctly forecasts rain 90% of the time. When it doesn't rain, he incorrectly forecasts rain 10% of the time. What is the probability that it will rain on Marie's wedding?

Bayes Rule example

Marie is getting married tomorrow, at an outdoor ceremony in the desert. In recent years, it has rained only 5 days each year (5/365 = 0.014). Unfortunately, the weatherman has predicted rain for tomorrow. When it actually rains, the weatherman correctly forecasts rain 90% of the time. When it doesn't rain, he incorrectly forecasts rain 10% of the time. What is the probability that it will rain on Marie's wedding?

$$P(\text{Rain}) = 0.014$$
$$P(\text{Predict} \mid \text{Rain}) = 0.9$$
$$P(\text{Predict} \mid \neg\text{Rain}) = 0.1$$
$$P(\text{Rain} \mid \text{Predict}) = ?$$

Bayes Rule example

Marie is getting married tomorrow, at an outdoor ceremony in the desert. In recent years, it has rained only 5 days each year (5/365 = 0.014). Unfortunately, the weatherman has predicted rain for tomorrow. When it actually rains, the weatherman correctly forecasts rain 90% of the time. When it doesn't rain, he incorrectly forecasts rain 10% of the time. What is the probability that it will rain on Marie's wedding?

$$
\begin{aligned}
P(\text{Rain}|\text{Predict}) &= \frac{P(\text{Predict}|\text{Rain})P(\text{Rain})}{P(\text{Predict})} \\
&= \frac{P(\text{Predict}|\text{Rain})P(\text{Rain})}{P(\text{Predict}|\text{Rain})P(\text{Rain}) + P(\text{Predict}|\neg\text{Rain})P(\neg\text{Rain})} \\
&= \frac{0.9 \times 0.014}{0.9 \times 0.014 + 0.1 \times 0.986} = 0.111
\end{aligned}
$$

Independence

Two events A and B are independent if and only if
$P(A \wedge B) = P(A)\, P(B)$

- In other words, $P(A \mid B) = P(A)$ and $P(B \mid A) = P(B)$
- This is an important simplifying assumption for modeling, e.g., *Toothache* and *Weather* can be assumed to be independent

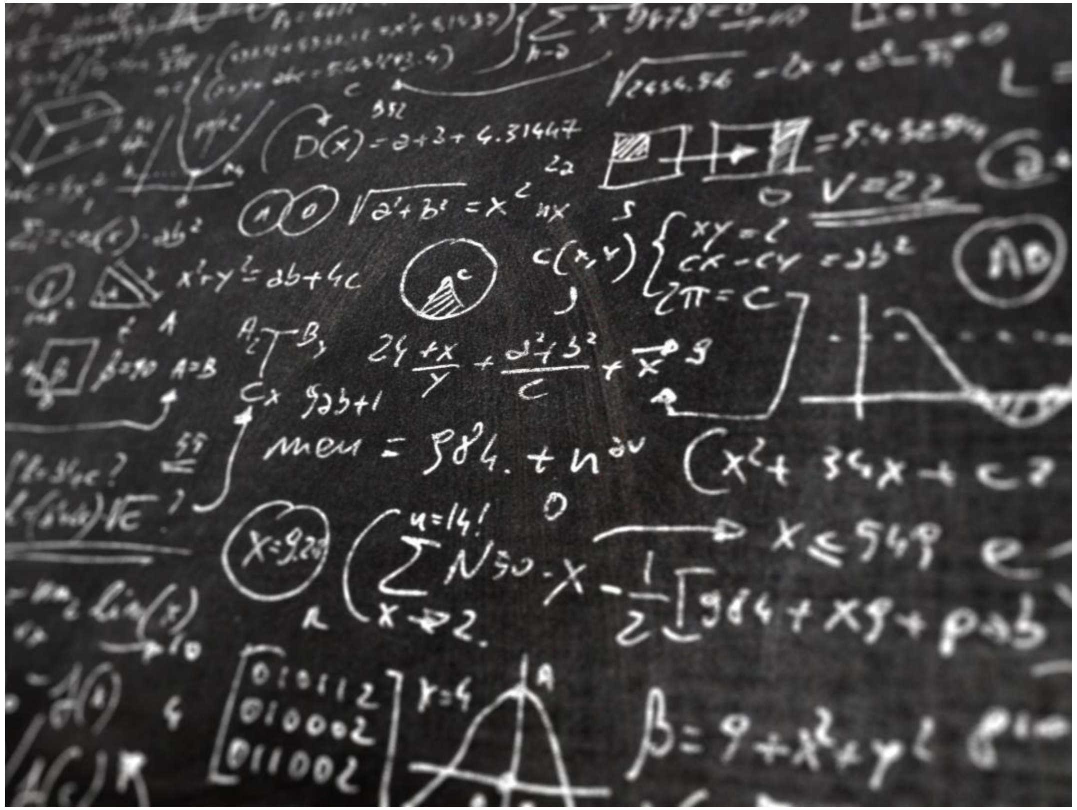
Are two ***mutually exclusive*** events independent?

- No, but for mutually exclusive events we have
  $P(A \wedge B) = 0$, $P(A \vee B) = P(A) + P(B)$

Probabilistic inference

In general, the **agent** observes the values of some **random variables** $X_1, X_2, \ldots, X_n$ and needs to **reason** about the values of some other ***unobserved*** **random variables** $Y_1, Y_2, \ldots, Y_m$

- Figuring out a diagnosis based on symptoms and test results
- Classifying the content type of an image or a document based on some features



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Bayesian inference as a Classifier

Each attribute and Class is considered a random variable

$(A_1, A_2, \ldots, A_n)$ Let a record with attributes be given

- The goal is to find the class C to which this record belongs.
- Specifically, it is the task of finding the class C that will maximize the probability of P(C| A1, A2,…,An).

Can we find the probability value $P(C| A_1, A_2, \ldots, A_n)$ from the data itself?

Naïve Bayes Classifier

Assumes independence among A_i attributes:

- $P(A_1, A_2, \ldots, A_n | C) = P(A_1 | C_j)\, P(A_2 | C_j) \ldots P(A_n | C_j)$

- Thus, for each value of A_i and C_j, the probability of $P(A_i | C_j)$ is found.

- Each new point is allocated to the class C_j, maximizing its probability, $P(C_j)$, $P(A_i | C_j)$.

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Naïve Bayes Classifier Example

For given

Test record;

$$X = (\text{Refund} = \text{No}, \text{Married}, \text{Income} = 120\text{K})$$

naive Bayes Classifier:

P(Refund=Yes|No) = 3/7
P(Refund=No|No) = 4/7
P(Refund=Yes|Yes) = 0
P(Refund=No|Yes) = 1
P(Marital Status=Single|No) = 2/7
P(Marital Status=Divorced|No)=1/7
P(Marital Status=Married|No) = 4/7
P(Marital Status=Single|Yes) = 2/7
P(Marital Status=Divorced|Yes)=1/7
P(Marital Status=Married|Yes) = 0

For taxable income:

If class=No:	sample mean=110
	sample variance=2975
If class=Yes:	sample mean=90
	sample variance=25

P(X|Class=No) = P(Refund=No|Class=No)
$*$ P(Married| Class=No)
$*$ P(Income=120K| Class=No)
= 4/7 $*$ 4/7 $*$ 0.0072 = 0.0024

P(X|Class=Yes) = P(Refund=No| Class=Yes)
$*$ P(Married| Class=Yes)
$*$ P(Income=120K| Class=Yes)
= 1 $*$ 0 $*$ 1.2 $*$ 10^{-9} = 0

P(X|No)P(No) > P(X|Yes)P(Yes)

Thus, P(No|X) > P(Yes|X)

=> Class = No

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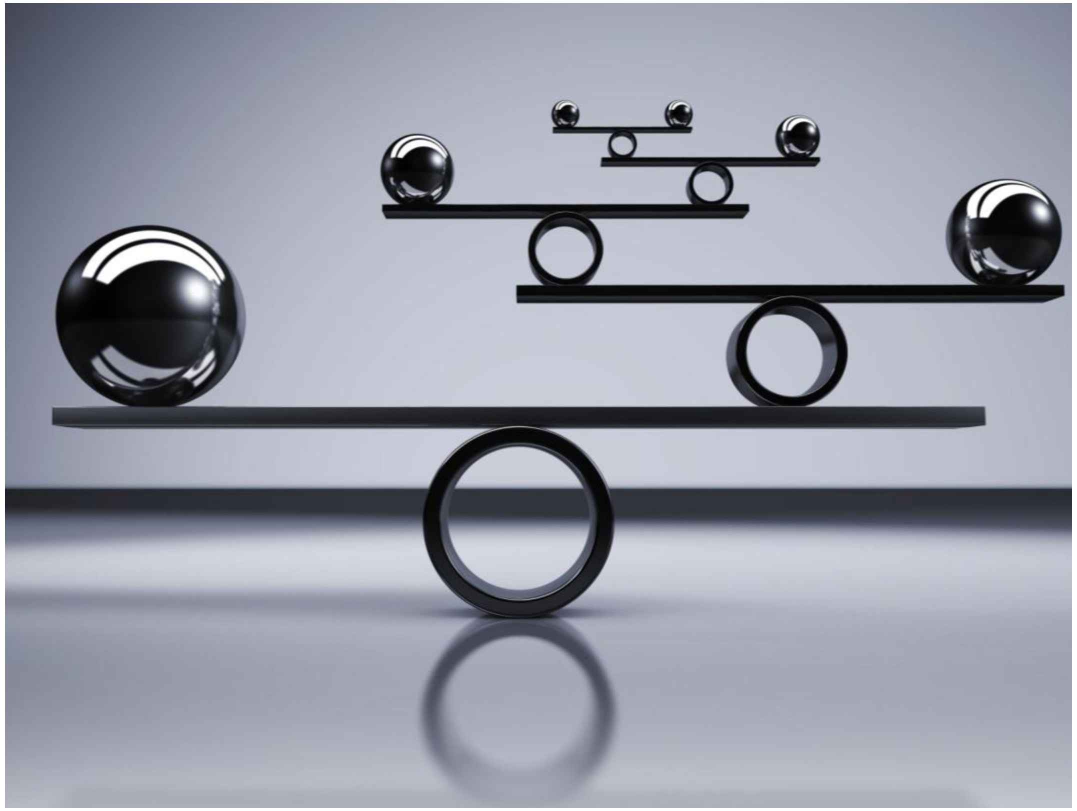
What if variables are not independent?

- **Assumption of independence** is not true for some attributes
- Many variables in real-world are **causal** or **correlated**



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Probabilistic inference

Suppose you are trying to determine if a patient has inhalational anthrax. You observe the following symptoms:

- The patient has a cough
- The patient has a fever
- The patient has difficulty breathing

Reasoning with uncertainty

You would like to determine how likely the patient is infected with inhalational anthrax given that the patient has a cough, a fever, and difficulty breathing

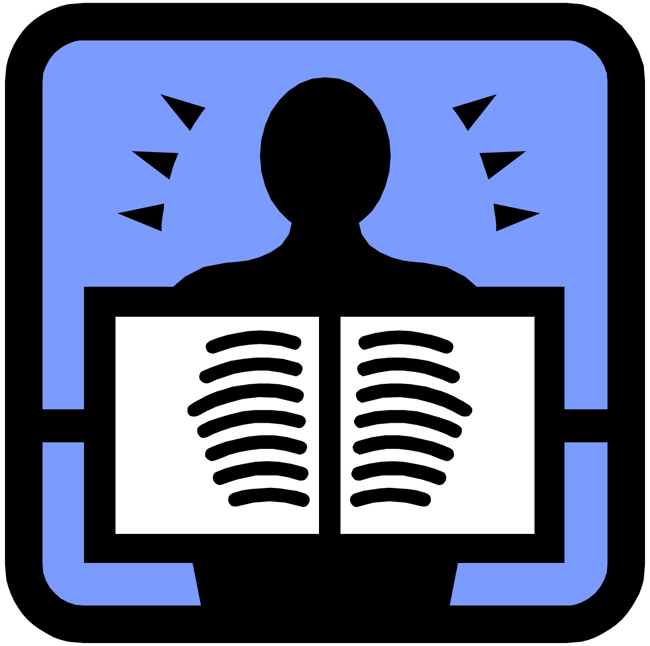
We are not 100% certain that the patient has anthrax because of these symptoms. We are dealing with uncertainty!

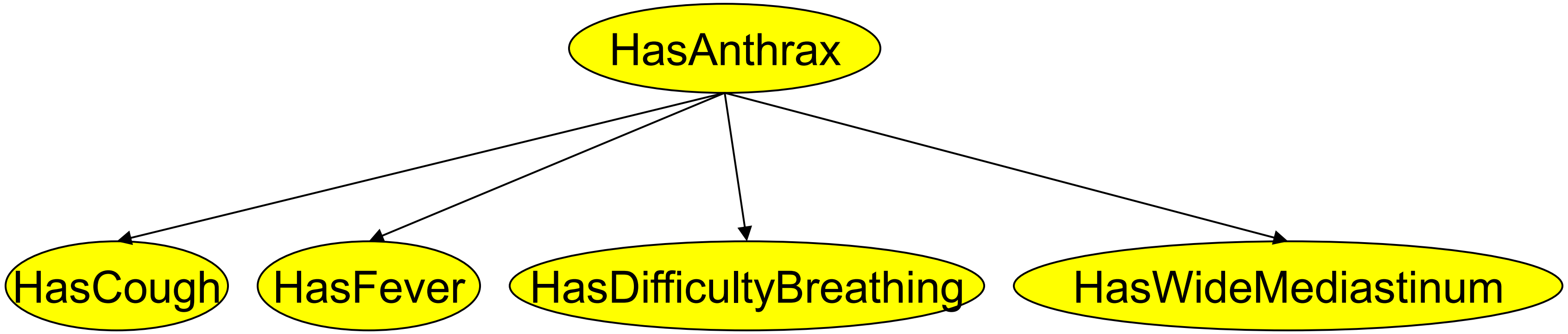
Reasoning with uncertainty

Now suppose you order an x-ray and observe that the patient has a wide mediastinum.

Your belief that the patient is infected with inhalational anthrax is now much higher.

- **Observations affected** your **belief** that the patient is infected with anthrax
- Reasoning with uncertainty
- Need **methodology** for **reasoning with uncertainty**...





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The Problem with the Joint Distribution

- **Lots of entries in the table to fill up!**
- **For *k* Boolean random variables, you need a table of size 2^k**
- **How do we use fewer numbers?**

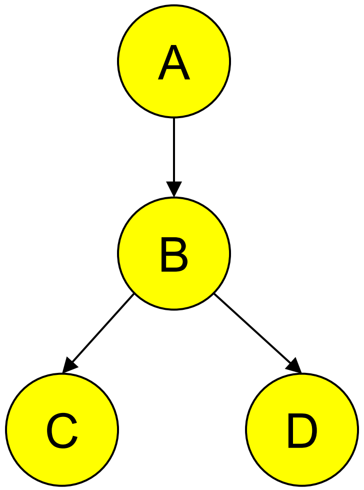
A	B	C	P(A,B,C)
false	false	false	0.1
false	false	true	0.2
false	true	false	0.05
false	true	true	0.05
true	false	false	0.3
true	false	true	0.1
true	true	false	0.05
true	true	true	0.15

A Bayesian Network

A Bayesian network (also called a belief network) consists of:

1. A **Directed Acyclic Graph**

   No loops of any length are allowed.



2. A set of **tables** for **each node** in the graph

A	P(A)
false	0.6
true	0.4

A	B	P(B|A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	D	P(D|B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

- Bayesian networks are one of the most significant contribution in AI in the last 20-40 years
- They are used in many applications, e.g., spam filtering, speech recognition, robotics, diagnostic systems and even syndromic surveillance

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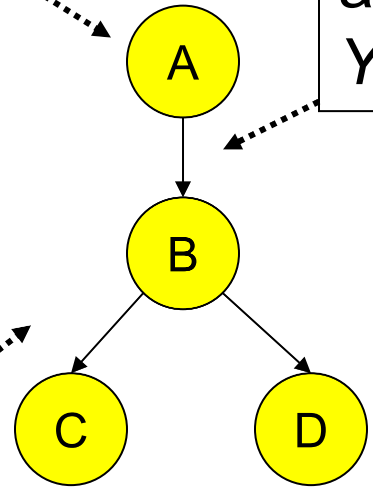
Bayesian Network as a Directed Acyclic Graph

Each node in the graph is a random variable

A node *X* is a parent of another node *Y* if there is an arrow from node *X* to node *Y* eg. *A* is a parent of *B*



Informally, an arrow from node *X* to node *Y* means *X* has a direct influence on *Y*

Two important properties:

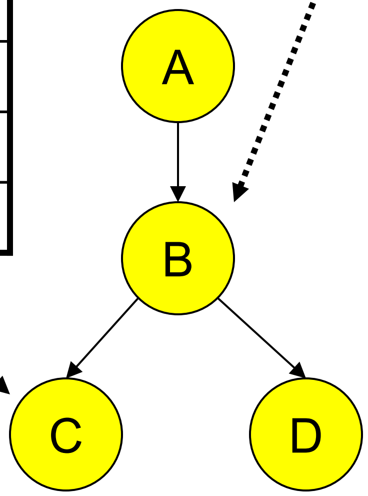
1. Encodes the **conditional independence** relationships between the variables in the **graph structure**
2. Is a **compact representation** of the **joint probability distribution** over the variables

A Set of Tables for Each Node

A	P(A)
false	0.6
true	0.4

A	B	P(B|A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1



B	D	P(D|B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

Each node X_i has a conditional probability distribution P(X_i | Parents(X_i)) that quantifies the effect of the parents on the node

The parameters are the probabilities in these conditional probability tables (CPTs)

A Set of Tables for Each Node

Conditional Probability
Distribution for C given B

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

For a given combination of values of the parents (B in this example), the entries for P(C=true | B) and P(C=false | B) must add up to 1,
eg. P(C=true | B=false) + P(C=false | B=false)=1

If you have a Boolean variable with k Boolean parents, this table has 2^{k+1} probabilities

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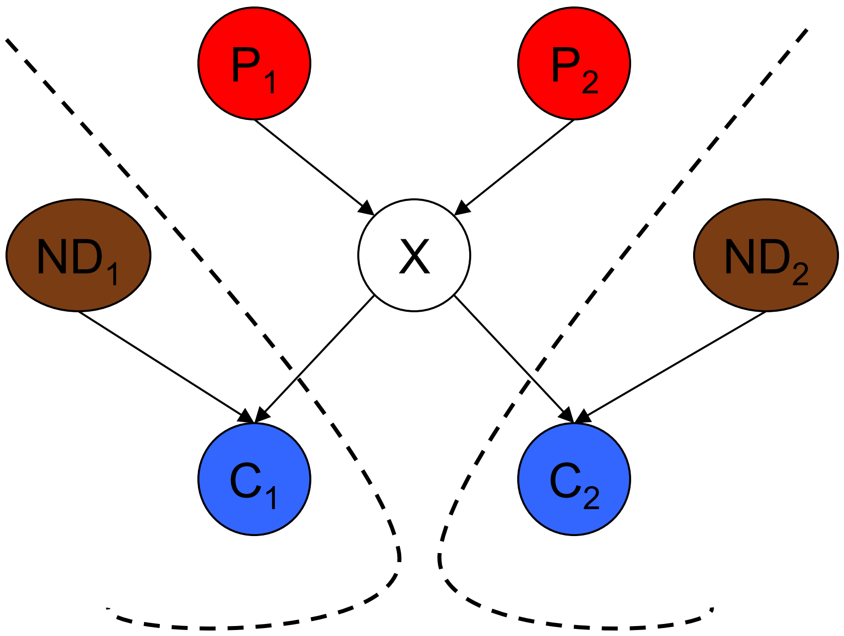
Conditional Independence

The Markov condition: given its parents (P_1, P_2), a node (X) is **conditionally independent** of its non-descendants (ND_1, ND_2)



Conditional Independence: Example

A is **the height of a child** and **B** is **the number of words that the child knows**.

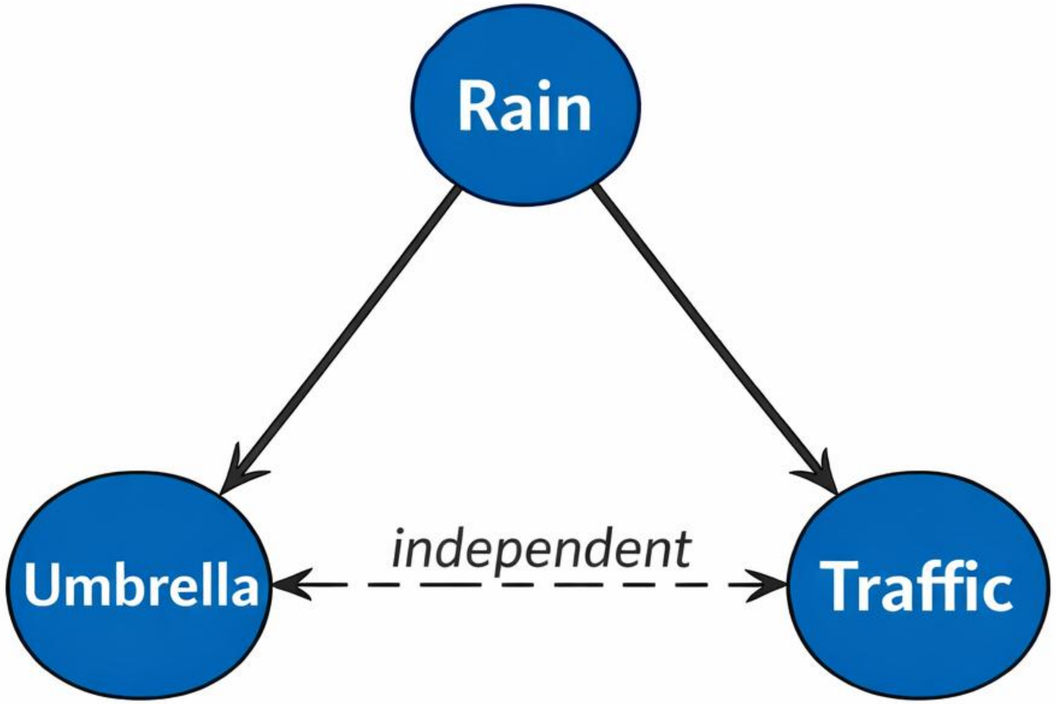
- When **A** is high, **B** is high too.

A **single piece of information will make A and B completely independent.**

- The child's age.

The height and the # of words known by the kid are **NOT independent**, but they are **conditionally independent** if the kid's age is provided.

Conditional Independence Example



Relationship:

- Rain affects both umbrella usage and traffic
- But umbrella and traffic are **conditionally independent** given **rain**

- R: Rain
- U: Umbrella
- T: Traffic

Conditional independence: $U \perp T \mid R$

$$P(U, T \mid R) = P(U \mid R)\, P(T \mid R)$$

Equivalent form:

$$P(U \mid T, R) = P(U \mid R)$$

The Joint Probability Distribution

Due to the **Markov condition**, we can compute the **joint probability distribution** over all the variables $X_1, \ldots, X_n$ in the **Bayesian net** using the formula:

Chain rule: $$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i | X_1 = x_1, X_2 = x_2, \ldots X_{i-1} = x_{i-1}))$$

Conditional independence: $$P(X_i = x_i | X_1 = x_1, X_2 = x_2, \ldots X_{i-1} = x_{i-1})) = P(X_i = x_i | Parents(X_i))$$

->Consequence: $$P(X_1 = x_1, \ldots, X_n = x_n) = \prod_{i=1}^{n} P(X_i = x_i | Parents(X_i))$$

$\boxed{\text{Where Parents}(X_i)\text{ means the values of the Parents of the node }X_i\text{ with respect to the graph}}$

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Inference on Bayesian Network

- Using a Bayesian network to compute probabilities is called inference

- In general, inference involves queries of the form:
  $P(X \mid E)$

E = The evidence variable(s)

X = The query variable(s)

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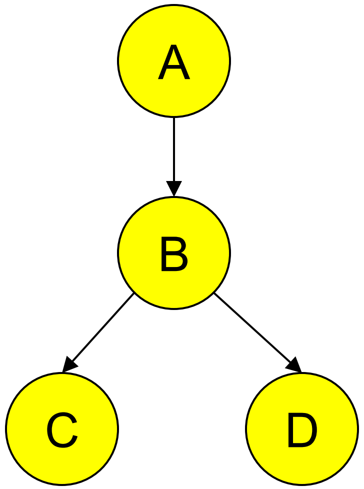
Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

P(A = true, B = true, C = true, D = true)
= P(A = true) * P(B = true | A = true) *
P(C = true | B = true) * P(D = true | B = true)
= (0.4)*(0.3)*(0.1)*(0.95)



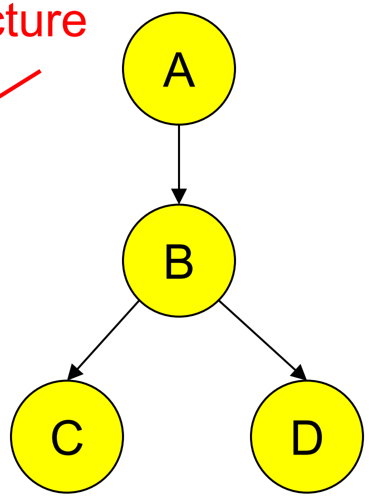
Using a Bayesian Network Example

Using the network in the example, suppose you want to calculate:

This is from the graph structure

$$
\begin{aligned}
&P(A = \text{true}, B = \text{true}, C = \text{true}, D = \text{true})\\
&= P(A = \text{true}) * P(B = \text{true} \mid A = \text{true}) * P(C = \text{true} \mid B = \text{true}) * P(D = \text{true} \mid B = \text{true})\\
&= \underbrace{(0.4)*(0.3)*(0.1)*(0.95)}
\end{aligned}
$$

These numbers are from the conditional probability tables



A	P(A)
false	0.6
true	0.4

A	B	P(B|A)
false	false	0.01
false	true	0.99
true	false	0.7
true	true	0.3

B	C	P(C|B)
false	false	0.4
false	true	0.6
true	false	0.9
true	true	0.1

B	D	P(D|B)
false	false	0.02
false	true	0.98
true	false	0.05
true	true	0.95

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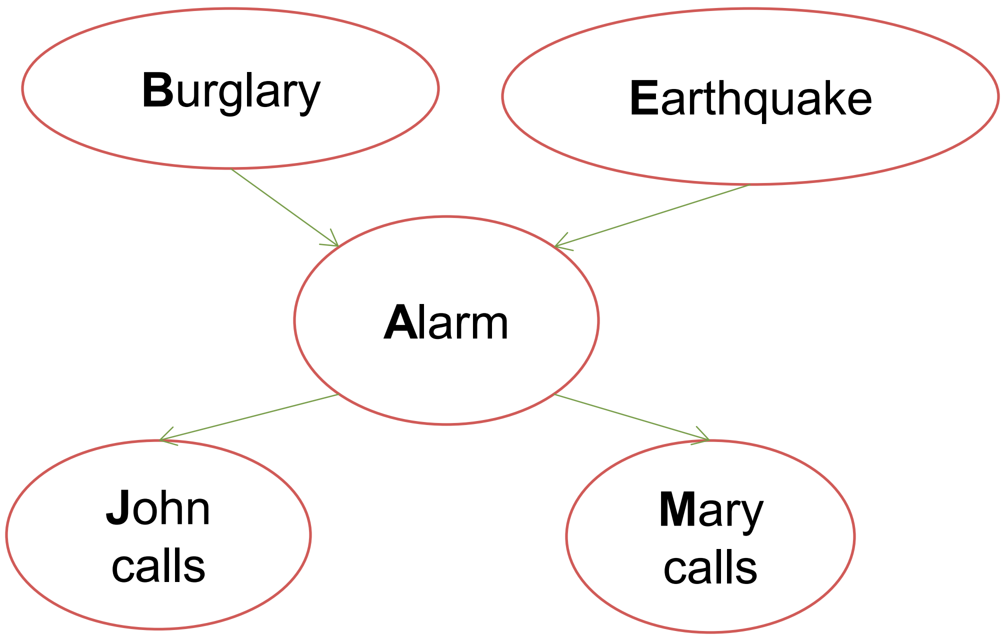
Example: Alarm Network

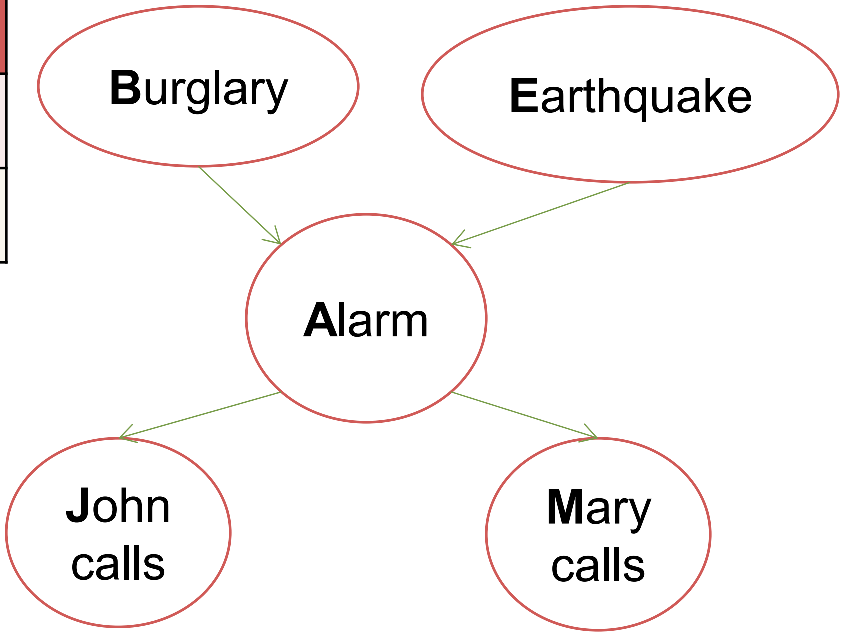


* Mary & John are my neighbors

How many parameters?

Example: Alarm Network

B	P(B)
+b	0.001
¬b	0.999



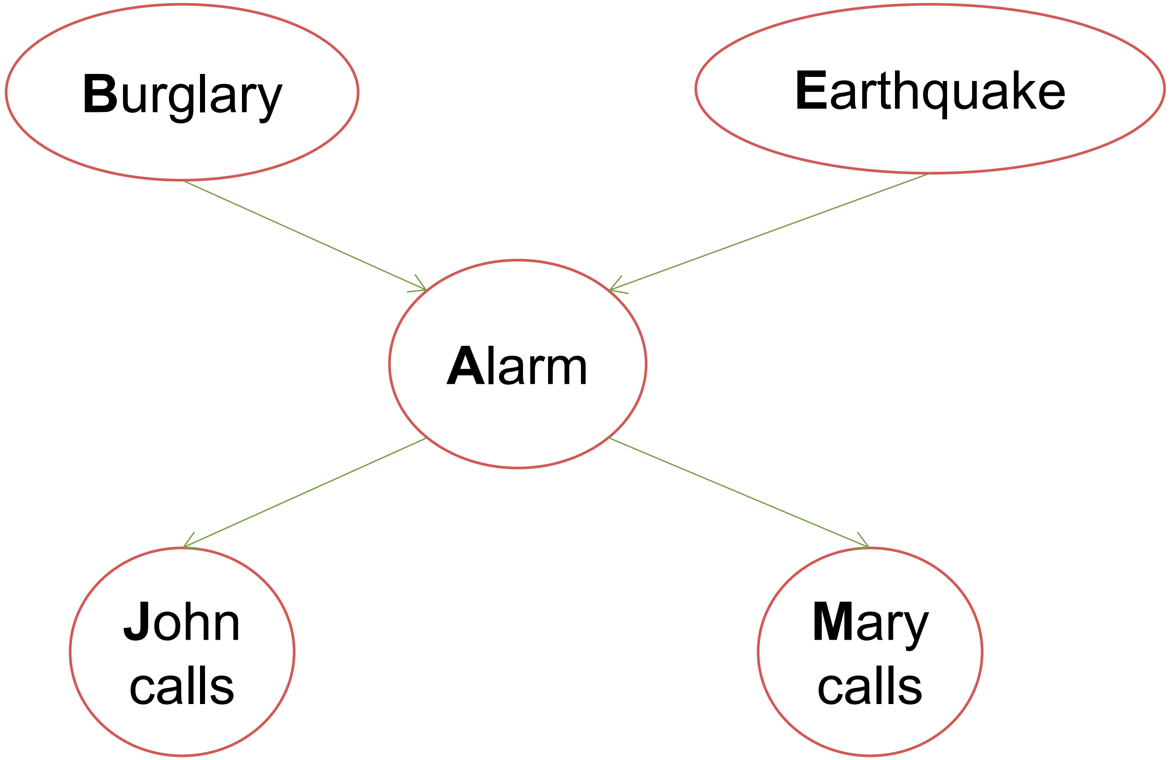
E	P(E)
+e	0.002
¬e	0.998

B	E	A	P(A|B,E)
+b	+e	+a	0.95
+b	+e	¬a	0.05
+b	¬e	+a	0.94
+b	¬e	¬a	0.06
¬b	+e	+a	0.29
¬b	+e	¬a	0.71
¬b	¬e	+a	0.001
¬b	¬e	¬a	0.999

A	J	P(J|A)
+a	+j	0.9
+a	¬j	0.1
¬a	+j	0.05
¬a	¬j	0.95

A	M	P(M|A)
+a	+m	0.7
+a	¬m	0.3
¬a	+m	0.01
¬a	¬m	0.99

Example: Alarm Network



$$\prod_i P(X_i|\text{Parents}(X_i)) = P(B) \cdot P(E) \cdot P(A|B,E) \cdot P(J|A) \cdot P(M|A)$$

$$P(+b, -e, +a, -j, +m) = P(+b)P(-e)P(+a|+b, -e)P(-j|+a)P(+m|+a) =$$

Building a Bayes Net

1. Choose a set of **relevant variables** and **an ordering** for them.
2. Assume they're called $X_1, \ldots, X_m$ (where X_1 is the first in the ordering, X_2 is the second, etc.)
3. For $i = 1$ *to* m:
   i. Add the X_i node to the network
   ii. Set *Parents(X_i)* to be a minimal subset of $\{X_1 \ldots X_{i-1}\}$ such that we have **conditional independence** of X_i and all other members of $\{X_1 \ldots X_{i-1}\}$ given *Parents(X_i)*
   iii. Define the **probability table** of

$$P(X_i = k \mid \text{Assignments of } Parents(X_i)\ ).$$

Example Bayes Net Building

T: The lecture started by 8:30

L: The lecturer arrives late

R: The lecture concerns robots

M: The lecturer is Me

S: It is sunny

- T only directly influenced by L (i.e., T is conditionally independent of R,M,S given L)
- L only directly influenced by M and S (i.e., L is conditionally independent of R given M & S)
- R only directly influenced by M (i.e., R is conditionally independent of L,S, given M)
- M and S are independent

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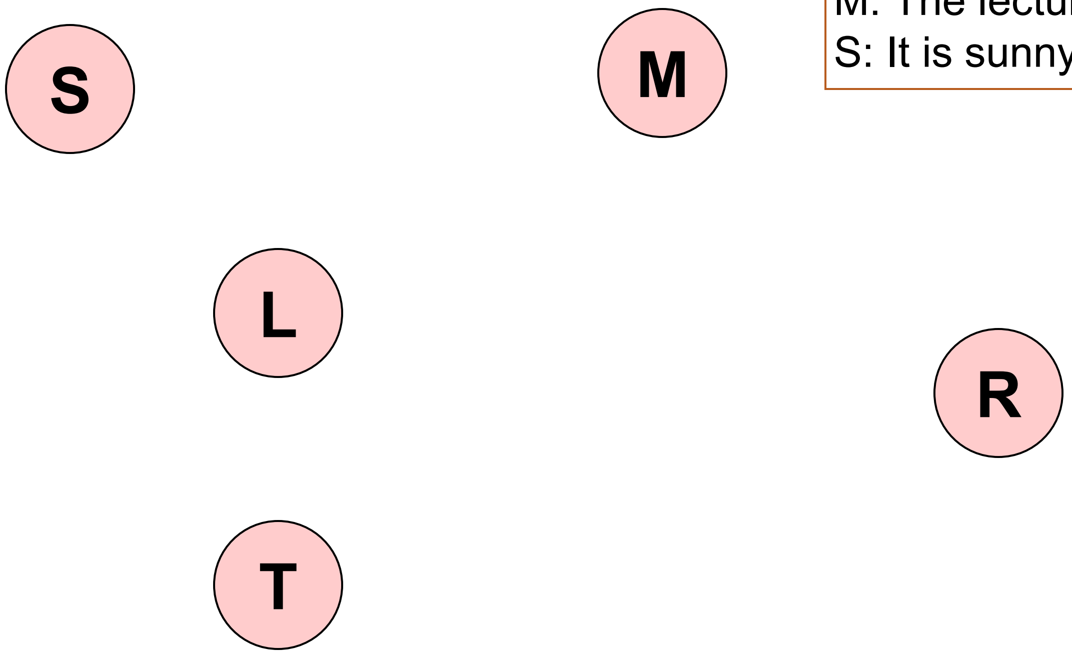
Making a Bayes net

T: The lecture started by 8:30
L: The lecturer arrives late
R: The lecture concerns robots
M: The lecturer is Me
S: It is sunny



Step One: add variables

- Just choose the variables you'd like to be included in the net.

Making a Bayes net

T: The lecture started by 8:30
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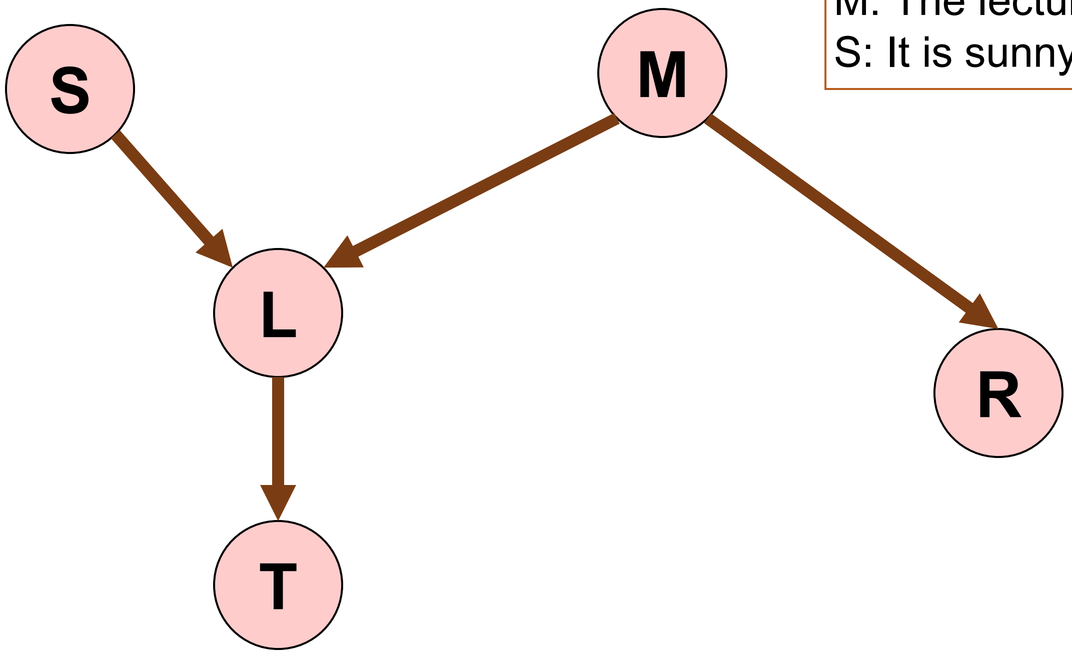

Step Two: add links

- The link structure must be acyclic (no cycles are allowed).
- If node X is given parents $Q_1,Q_2,..Q_n$ you are promising that any variable that's a non-descendent of X is conditionally independent of X given $\{Q_1,Q_2,..Q_n\}$

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Making a Bayes net

T: The lecture started by 8:30
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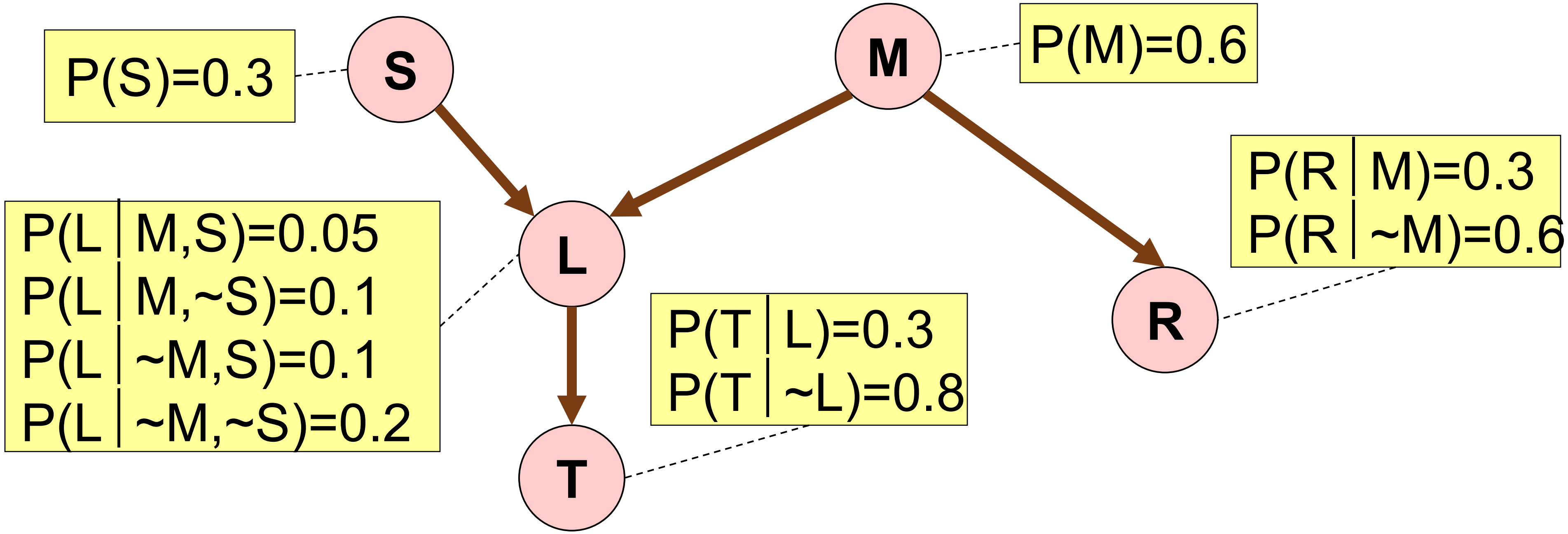
S: It is sunny

"~" : another symbol for "not"



Step Three: add a probability table for each node.

- The table for node X must list P(X|Parent Values) for each possible combination of parent values

Making a Bayes net

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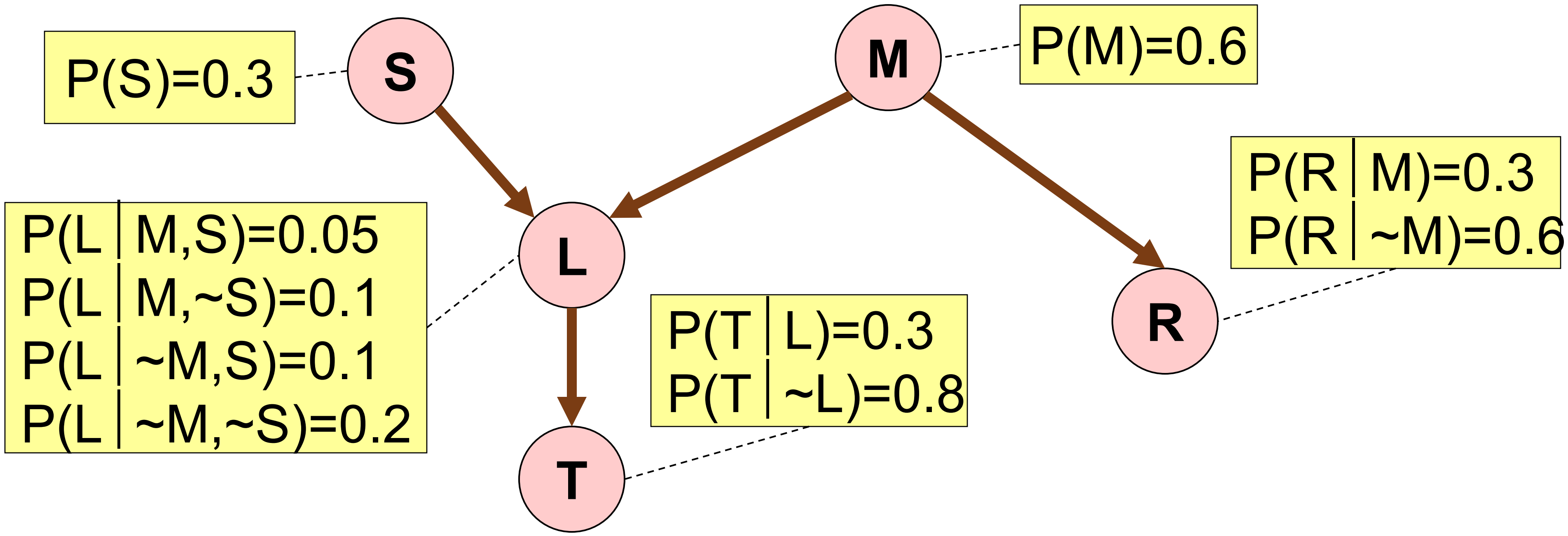

- Two **unconnected variables** may still be **correlated**
- Each node is **conditionally independent** of all **non-descendants** in the tree, **given its parents**.
- You can **deduce** many other **conditional independence** relations from a **Bayes net**.

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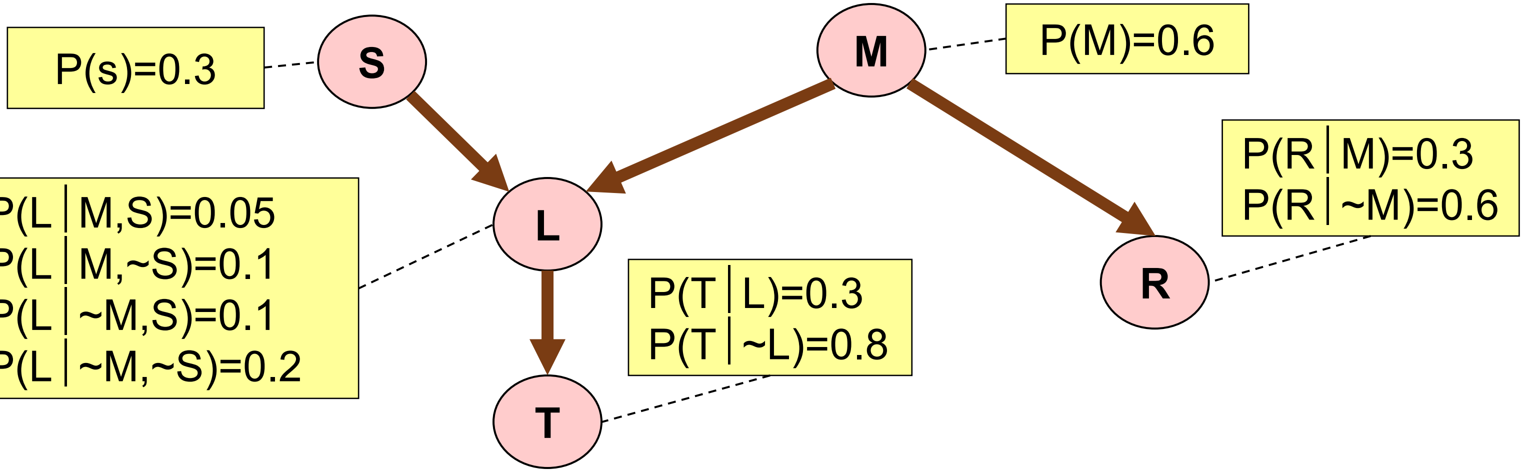
Computing a Joint Entry

How to compute an entry in a joint distribution?

E.G: What is P(S, ~M, L, ~R, T)?



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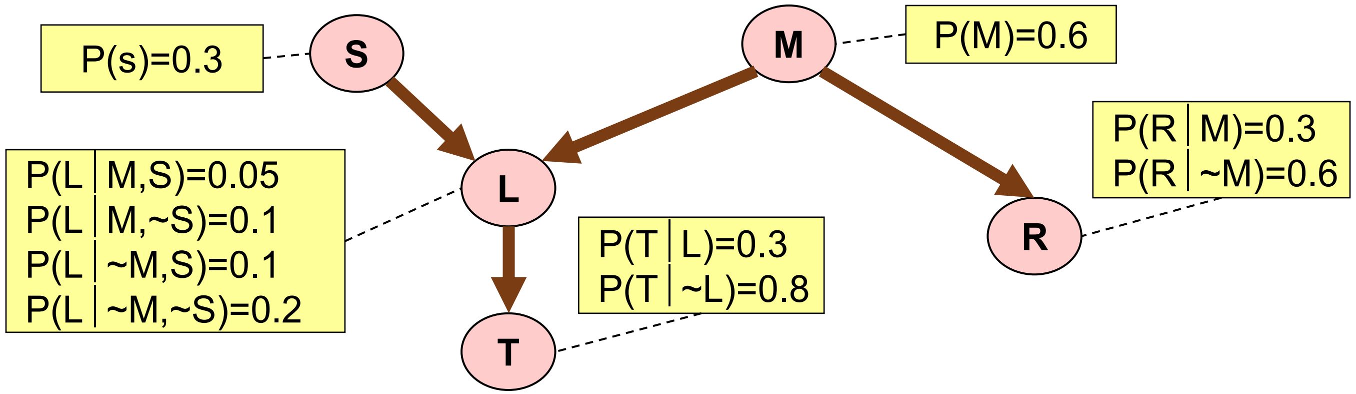
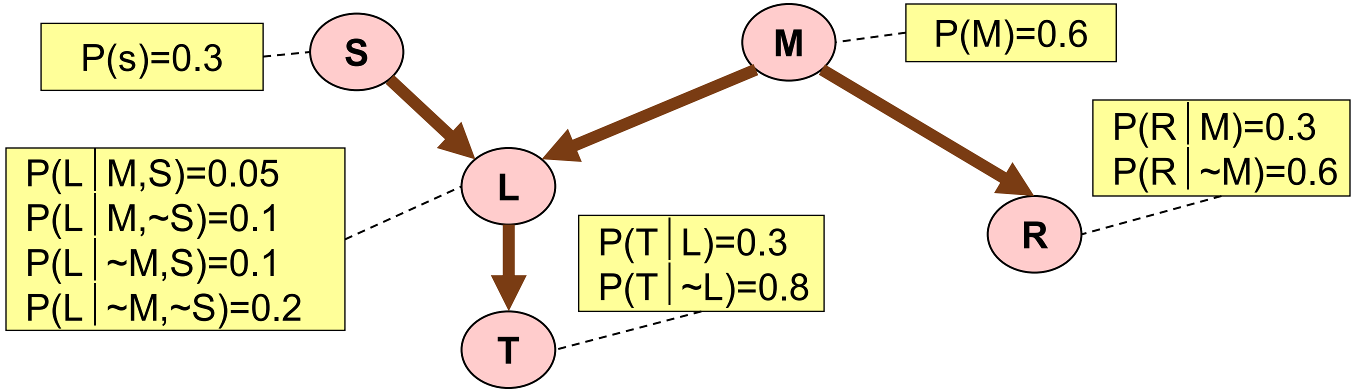
Computing with Bayes Net



P(T , ~R , L , ~M , S) =
= P(T | ~R , L , ~M , S) * P(~R , L , ~M , S) =
= P(T | L) * P(~R , L , ~M , S) =
= P(T | L) * P(~R | L , ~M , S) * P(L , ~M , S) =
= P(T | L) * P(~R | ~M) * P(L , ~M , S) =
= P(T | L) * P(~R | ~M) * P(L | ~M , S) * P(~M , S) =
= P(T | L) * P(~R | ~M) * P(L | ~M , S) * P(~M | S) * P(S) =
= P(T | L) * P(~R | ~M) * P(L | ~M , S) * P(~M) * P(S).

Where are we now?

- We have a methodology for building Bayes nets.
- **No exponential storage** to hold our probability table. Only **exponential** in the **maximum number of parents** of any node.
- We can compute **probabilities** in time **linear** with the number of **nodes**.
- So we can also **compute answers to any question**.



E.G. What could we do to compute P(R | T,~S)?

Where are we now?

Step 1: Compute P(R , T , ~S)

Step 2: Compute P(~R , T , ~S)

Step 3: Return

$$\frac{P(R, T, \sim S)}{P(R, T, \sim S) + P(\sim R, T, \sim S)}$$

g Bayes nets.

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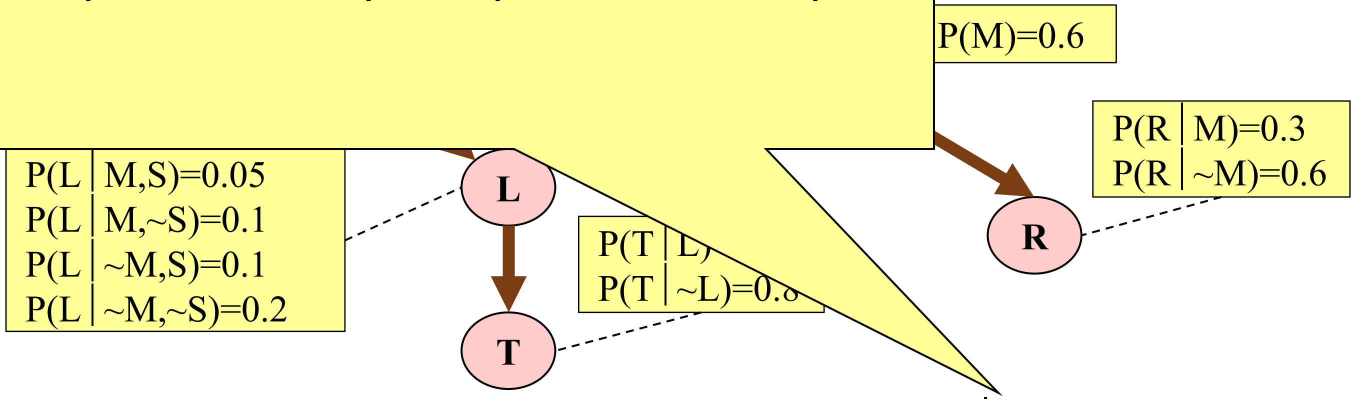
o any questions.



E.G. What could we do to compute P(R | T,~S)?

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Where are we now?

Step 1: Compute P(R , T , ~S)

Sum of all the rows in the Joint that match R , T , ~S

Step 2: Compute P(~R , T , ~S)

Sum of all the rows in the Joint that match ~R , T , ~S

Step 3: Return

$$\frac{P(R , T , \sim S)}{P(R , T , \sim S) + P(\sim R , T , \sim S)}$$

…ximum number of parents of

…ny

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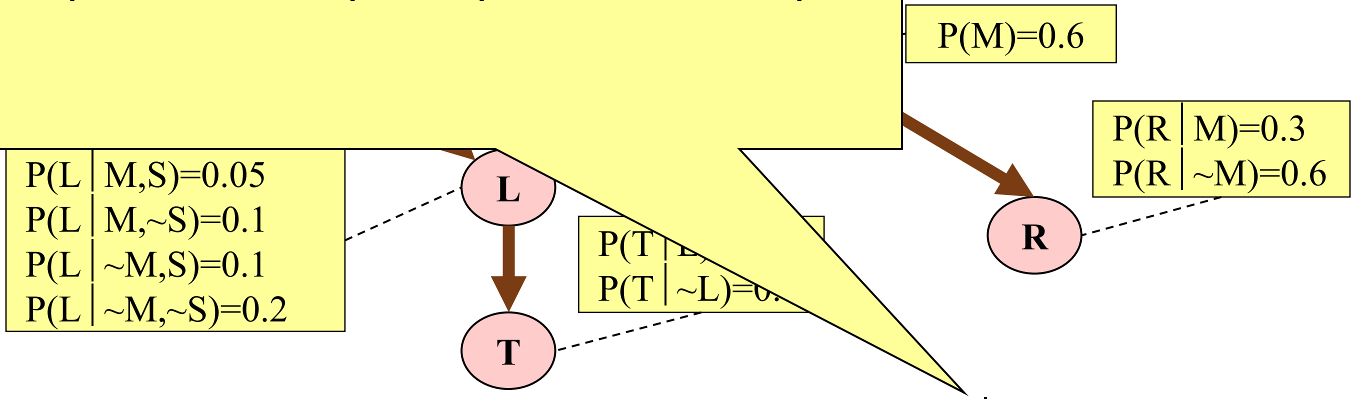
…to any questions.



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SDU

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Step 1: Compute P(R , T , ~S)

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Step 3: Return

$$\frac{P(R, T, \sim S)}{P(R, T, \sim S) + P(\sim R, T, \sim S)}$$

Sum of all the rows in the Joint that match R , T , ~S

4 joint computes

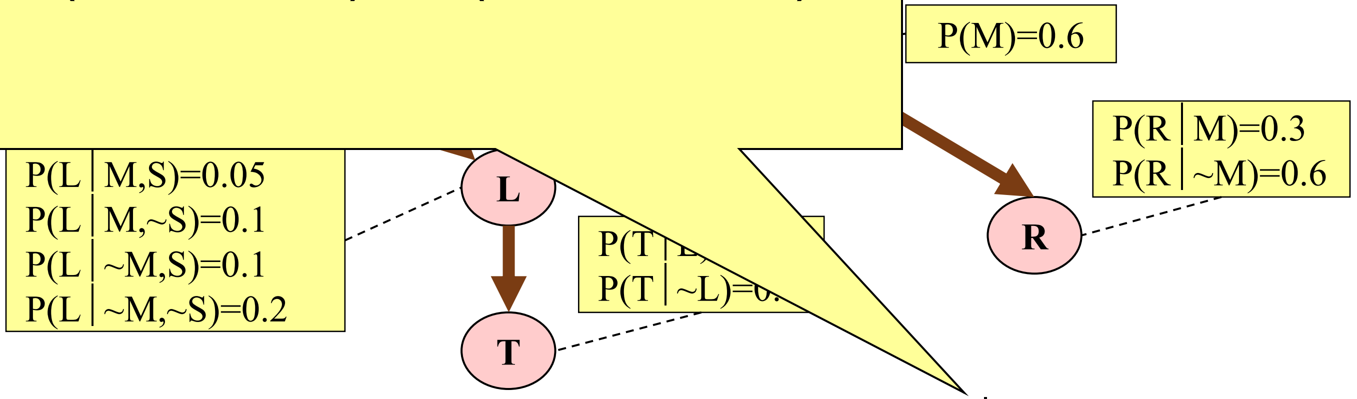
Sum of all the rows in the Joint that match ~R , T , ~S

4 joint computes

ximum number of parents of

an do it in time line... with the

to any questions.



E.G. What could we do to compute P(R | T,~S)?

The good news

We can do inference. We can compute any conditional probability:

P(Some variable | Some other variable values)

$$P(E_1 \mid E_2) = \frac{P(E_1 \wedge E_2)}{P(E_2)} = \frac{\sum_{\text{joint entries matching } E_1 \text{ and } E_2} P(\text{joint entry})}{\sum_{\text{joint entries matching } E_2} P(\text{joint entry})}$$

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The good news

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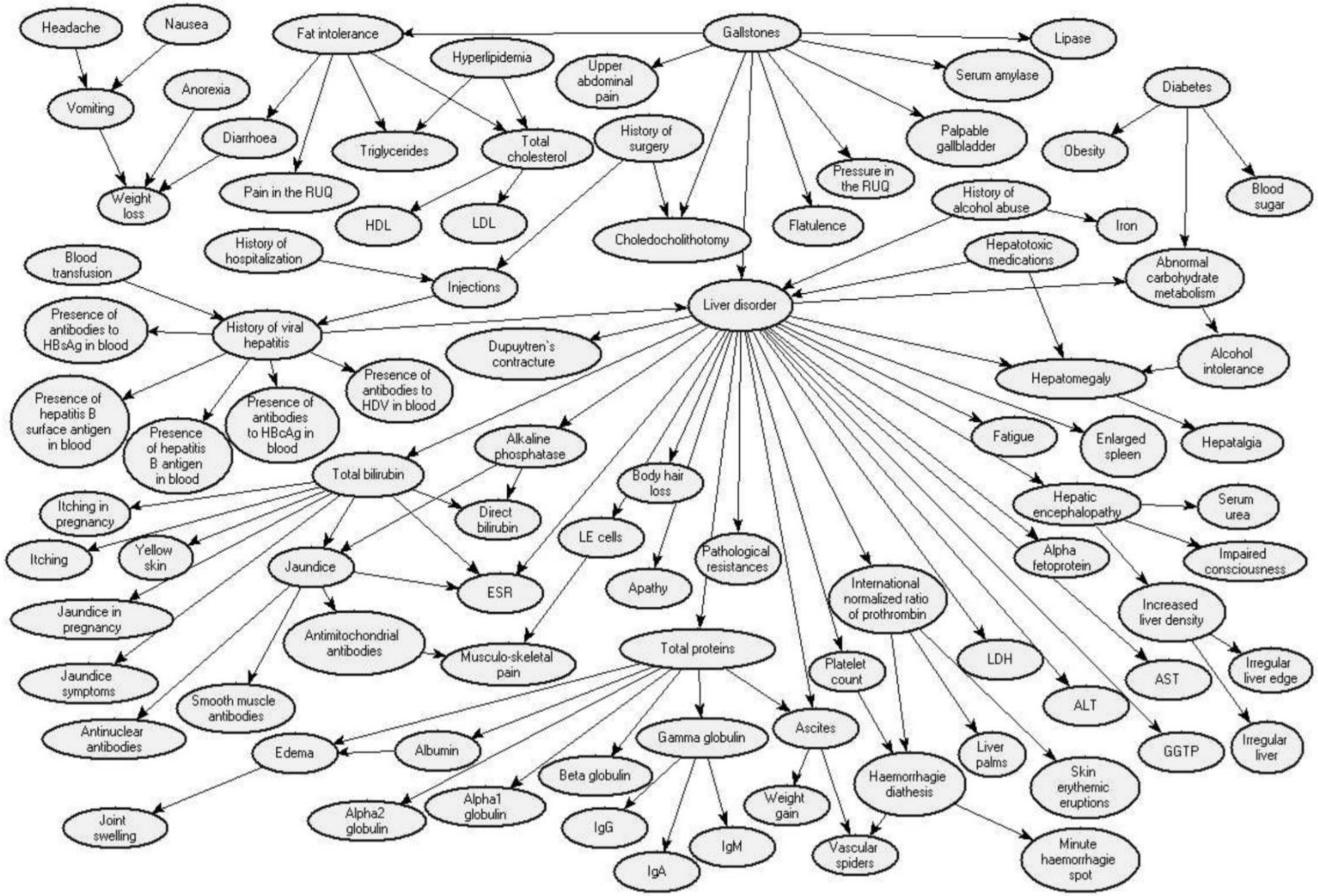
Suppose you have *m* binary-valued variables in your Bayes Net and expression E_2 mentions *k* variables.

How much work is the above computation?

Bayes Nets for Real-world problems can be large



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The sad, bad news

Conditional probabilities by enumerating all matching entries in the joint are expensive:

- **Exponential in the number of variables.**

But perhaps there are faster ways of querying Bayes nets?

- **Exact inference** is **feasible** in **small to medium-sized** networks
- Exact inference in **large networks** takes a **very long time**
- We resort to **approximate inference techniques** which are much **faster** and give pretty good results
- If ever asked to manually do a Bayes Net inference -> many tricks to save you time -> not topic of this class though ☹

Sadder and worse news: **General querying of Bayes nets is NP-complete.**

One last unresolved issue…

Where do we get the Bayesian network from?

Two options:

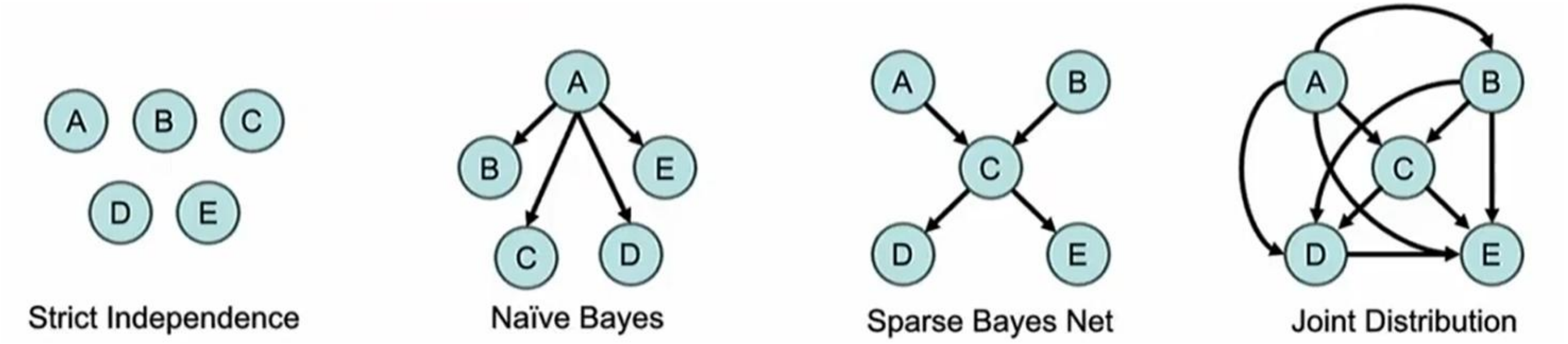
- Get an expert to design it
- Learn it from data

Summary

- **Independence** and **Conditional independence** are important forms of probabilistic knowledge

- **Bayes nets encode joint distributions efficiently** by taking advantage of conditional independence

- Allows flexible **trade off** between **model accuracy** and **compute efficiency**



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Thank you for your attention